

Abhishek Enaguthi

enaguthiabhishek@gmail.com 971-506-5811 linkedin/abhishekenaguthi github/Abhishek21g enaguthi.com

Summary

CPU architecture and systems engineer with hands-on experience in compiler backends, low-level debug, and HPC performance optimization.

Skills

Languages: Python, C, C++, Rust, Typescript, Java, JavaScript, SQL, Go

Frameworks: LLVM, CUDA, SLURM, PostgreSQL, REST APIs, Pytorch, Llama Index, RAG

Developer Tools: CI/CD, AWS, Lambda, Docker, Git, Figma, pyTest, Kubernetes

Experience

LLM Research Internship

June 2025 – September 2025

University of Washington

Seattle, Washington

- Identified gaps in RAG training data and developed multi-turn synthetic QA generation algorithms for NVIDIA NeMo
- Deployed LLaMA-3 70B, Mistral inference pipelines via vLLM on NVIDIA H200 clusters using Kubernetes and NCCL
- Delivered production RAG pipelines on a 50-node, 20 TB distributed system for 20+ staff

Systems Software Engineer

June 2024 – Present

College of Earth Science at OSU

Corvallis, Oregon

- Presented a physics-based digital twin of data centers at Supercomputing '24 using NVIDIA Omniverse and Isaac Sim
- Developed LLVM passes, kernel optimizations, custom backends boosting ML and data processing performance by 30%
- Built state coordination and orchestration pipelines over IBM LSF for job submission, retry logic, and log capture
- Debugged performance and reliability issues across Linux and Kubernetes env. for long-running engineering workflows

HPC Performance Engineer

March 2024 – Present

Oregon State University

Corvallis, Oregon

- Automated Slurm RPC and disk-usage monitoring with Bash scripts, adopted by 100+ users daily across OSU's HPC
- Achieved 20% faster collective communication and 15% higher memory throughput benchmarking NCCL and MPI
- Managed large-scale parallel storage systems including Lustre and GPFS, optimizing I/O throughput for ML workloads
- Developed Python scripts to automate DL and HPC testing and built tooling to visualize large-scale performance data

AI Research Engineer

March 2024 – Present

Prof. Christopher Swillan — Research Advanced Computing

Corvallis, Oregon

- Secured \$ 1M joint research grant from AMD and IBM to advance compiler technology for heterogeneous GPU systems
- Published compiler research enabling ROCm cross-compilation to x86 machines in collaboration with AMD and IBM
- Led a team to Supercomputing '25, placing top 3 out of 50+ teams in the IndySCC student cluster competition
- Presented physics-based digital twin research at NVIDIA GTC '25 to a global audience of researchers and engineers

Projects

Retrieval Augmented Generation (RAG) pipeline | Python, PyTorch, FAISS, MXBAI Model

May 2026

- Scaled embedding generation to 235K embeddings from 65K Wikipedia articles using PyTorch DDP across multi-GPU nodes

Mocki - Multi-agent interview coaching | React, Node.js, NVIDIA Nemotron, NIM API, Tailwind CSS

May 2026

- Built a multi-agent AI mock interview platform at BeaverHacks 2026, winning 1st place in the NVIDIA Nemotron track (RTX 5070 prize) out of 50+ teams
- Architected a 4-agent pipeline (Coordinator → Interviewer → Evaluator → Report Generator) using NVIDIA NIM API with adaptive difficulty across interview rounds

Education

Master of Science (Thesis option) in Artificial Intelligence

Expected June 2026

Oregon State University

GPA - 4.0

Bachelor of Science in Computer Science (AI), Minor in Finance

Expected June 2026

Oregon State University

GPA - 3.95

Relevant Coursework: Computer Architecture, OS I & II, Distributed Systems, Machine & Deep Learning

Leadership & Extracurriculars

- OSU Engineering Student Council — **VP of Development** Feb 2024 – Present
- Dean's List 7x, **2x Hackathon Winner**
- Published** a paper on **Digital Twin** for data centers at **SC24**
- Co-authoring** a paper for **ICML 2026** submission with two **UC Berkeley** collaborators